

Problem Set 3

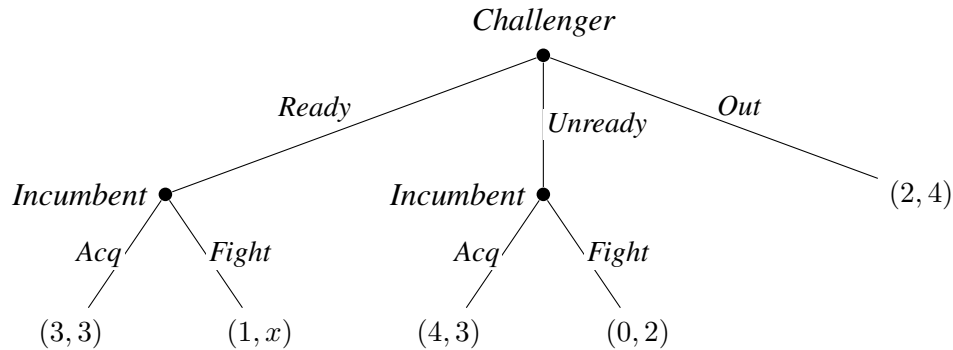
POLS W4732

Fall 2026

Due October 16 before the TA section

Artificial-intelligence policy. Students may not use generative-AI systems (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, and collaboration within the registered problem-set group are permitted. **Every group member is responsible for understanding and being able to explain the entire submission.**

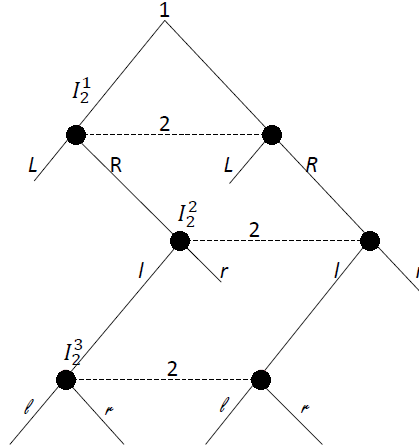
Exercise 1 (Entry and strategic-form representation). *Consider the game below.*



The incumbent payoff at (Ready, Fight) is the parameter x .

1. Construct the strategic form and list all Nash equilibria depending on x . For each, state its induced path.
2. List all SPE depending on x .
3. Is it fair to interpret x as the Incumbent's deterrence ability?

Exercise 2. Consider the following game, where I_2^1, I_2^2, I_2^3 denote the information sets for Pl. 2. Let $s_2 \equiv (L, l, r)$ and $s'_2 \equiv (R, r, l)$, denote two pure strategies for Pl. 2. Find the behavioral strategy equivalent to the mixed strategy that puts equal probability (i.e., $1/2$ and $1/2$) on s_2 and s'_2



Exercise 3. Consider the Public Good Provision Game, under the assumption that both costs are drawn from a uniform distribution in $[\gamma - \frac{1}{2\psi}, \gamma + \frac{1}{2\psi}]$, with $\gamma - (2\psi)^{-1} < 1 < \gamma + (2\psi)^{-1}$

1. Characterize the unique Nash Equilibrium
2. How does the equilibrium threshold c^* change with ψ and γ ? Can you explain these results intuitively?
3. How does the equilibrium probability that the public good is provided change with ψ and γ ? Can you briefly explain these results intuitively?
4. Now suppose that instead of two players, the game features n players. For simplicity, also assume that $\psi = 1/2$ and $\gamma = 1$. How does the equilibrium threshold c^* change with n ? Can you briefly articulate an intuition for this result?
5. (**hard**) Under the same assumptions as in the previous point, how does the probability that the public good is provided change with n (provided n is arbitrarily large)? Can you briefly articulate an intuition for this result (even if you are unable to show this)?

Exercise 4 (Incomplete-information war). Countries 1 and 2 choose F (mobilize) or N (accommodate). Mobilizing costs μ . If both mobilize, the winner receives $1 - \mu$ and the loser $-1 - \mu$; choosing F against N gives $-\mu$, and choosing N gives zero. Nature draws $\omega \in \{UR, DR, UL, DL\}$; Country 1 observes the first letter and Country 2 the second. Country 1 wins only under states UR and DL . The common prior satisfies $\Pr(UR) = \Pr(DL) = q/2$ and $\Pr(UL) = \Pr(DR) = (1 - q)/2$, with $q \in (0, 1)$ and $\mu \in (0, 1)$.

1. Construct a table of each type's posterior probability of winning.
2. Characterize all pure-strategy Bayesian equilibria as a function of (q, μ) .